THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 20	$E = 5F/2 = 30$; $V = 2 + E - F$
2	**(d)** 42	$5 \times 30 - 4 \times 27$
3	**(b)** 2	$9(a - b) = 9d$
4	**(d)** 768 cm²	$l + b = 64$, $l = 3b$: $b = 16$, $l = 48$
5	**(a)** 8	$V = 2E/3 = 12$; $F = 2 + E - V$
6	**(c)** 41	Correct sum $= 360 - 30 + 80$
7	**(a)** 7	$(x + 1/x)^2 = x^2 + 2 + 1/x^2$
8	**(a)** 128 cm²	Height $= \sqrt{10^2 - 6^2} = 8$ cm; area $= \frac{1}{2}(10 + 22) \times 8$
9	**(c)** 24	Each corner is replaced by 3 new vertices: $8 \times 3 = 24$
10	**(c)** 43	$(29 + 1)(14) - 29 \times 13$
11	**(d)** 7	$4d + 16$ = sum
12	**(c)** 1080	$1000 \times 120/100 \times 90/100$
13	**(a)** Both A and R are true and R is the correct explanation of A.	A cube has $F = 6$, $V = 8$, $E = 12$ and $6 + 8 - 12 = 2$; R is the general result.
14	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{4 + 6 + 8 + 10 + 12}{5} = \frac{40}{5} = 8$; R is the definition used.
15	**(a)** Both A and R are true and R is the correct explanation of A.	$2n$ is a multiple of 2 and $2n + 1 = 2n + 1$ leaves remainder 1; R is the definition behind the expressions.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$12(n - 2) = 48$ gives $n = 6$ (1); total $= 6^3 = 216$ (1).	1+1
17	Total $= 6 \times 20 = 120$ (1); rest $= 5 \times 18 = 90$, removed $= 30$ (1).	1+1
18	$a + b = 9$, $a - b = 3$ (1); $a = 6$, $b = 3$, number 63 (1).	1+1
19	$l + b = 50$, $l - b = 18$ (1); $l = 34$, $b = 16$, area $= 544$ cm² (1).	1+1
20	Each edge is shared by two faces: $E = (3 \times 8 + 4 \times 6)/2$ (1) $= 24$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$E = 3F/2 = 48$ (1); $V = 2 + E - F = 18$ (1); $F + V - E = 32 + 18 - 48 = 2$ (1).	1+1+1
22	New sum $= 2480 - 38 + 80 = 2522$ (2); mean $= 63.05$ (1).	2+1
23	$11(a + b) = 110$ so $a + b = 10$ (1); $a - b = 4$ gives $a = 7$, $b = 3$ (1); number $= 73$ (1).	1+1+1
24	$b = 20$ m, $l = 60$ m (1); area $= 1200$ m² (1); fencing cost $= 160 \times 15 =$ ₹ 2400 (1).	1+1+1
	OR: $12(n - 2) = 144$ gives $n = 14$ (1); one face: $6(n - 2)^2 = 6 \times 144 = 864$ (2).	1+2
25	Total age of players $= 253$ (1); total with coach $= 300$ (1); coach $= 47$ years (1).	1+1+1
	OR: $x^2 + 1/x^2 = 3^2 - 2 = 7$ (1); $x^4 + 1/x^4 = (7)^2 - 2 = 47$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $5 + 2 = 7$ (1). (ii) $2 \times 5 = 10$ (1). (iii) $3 \times 5 = 15$ (1). (iv) Prism: $F + V - E = 7 + 10 - 15 = 2$ (1). Pyramid: $F = 6$, $V = 6$, $E = 10$ and $6 + 6 - 10 = 2$ (1).	1+1+1+2
27	Using $(p + q)^2 - (p - q)^2 = 4pq$ (2); $4 \times 2n \times 3 = 24n$ (1); for $n = 4$: 96 (2).	2+1+2
	OR: Other diagonal $= 2 \times 216/18 = 24$ cm (2); side $= \sqrt{9^2 + 12^2} = 15$ cm (1); altitude $= 216/15 = 14.4$ cm (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 15	$2A/16$
2	**(a)** 256	$81 \times (4/3)^4$
3	**(d)** 40	Sort the 8 values; median is the mean of the 4th and 5th: 40
4	**(b)** 9	$a^2 - (a-3)(a+3) = 9$
5	**(b)** 840 cm²	Side $= 29$; other half-diagonal $= 21$; area $= \frac{1}{2} \times 40 \times 42$
6	**(b)** 4	Side halves every stage: $64/2^4$
7	**(c)** 48	Total distance 240 over total time
8	**(b)** 22	Middle integer is 21
9	**(d)** 1232 m²	$(52-8)(36-8)$
10	**(c)** 8	$E = (3 \times 8 + 4 \times 2)/2 = 16$; $V = 2 + E - F$
11	**(b)** 49	Mean is multiplied by k, then c is added
12	**(c)** 198	$(100a + 10b + c) - (100c + 10b + a) = 99(a - c)$
13	**(a)** Both A and R are true and R is the correct explanation of A.	$12 \times 5 = 60$ cm²; R is the formula used.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	A triangular prism has 2 triangular and 3 rectangular faces, so A is true; R is true but about pyramids, so it does not explain A.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	The middle value of the five ordered numbers is 9, so A is true; R is also true, but mode is a different measure and does not explain A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Sum $= 2 + 4 + \cdots + 16 = 72$ (1); mean $= 72 \div 8 = 9$ (1).	1+1
17	$x^2 - y^2 = (x+y)(x-y)$ (1) $= 85$ (1).	1+1
18	New area $= 100 \times 130/100 \times 80/100 = 104$ (1); change $= 4\%$ (1).	1+1
19	Number $= 3^5 = 243$ (1); side $= 128/2^5 = 4$ cm (1).	1+1
20	Wrong sum $= 198$ (1); correct sum $= 192$; mean $= 32$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Height $= \sqrt{15^2 - 9^2} = 12$ cm (2); area $= \frac{1}{2} \times (14 + 32) \times 12 = 276$ cm² (1).	2+1
22	$3^5 = 243 \le 500$ but $3^6 = 729 > 500$ (2); so stage 6 with 729 triangles (1).	2+1
23	Required total $= 6 \times 68 = 408$ (1); present total $= 310$ (1); needed $= 98$ (1).	1+1+1
24	$4d + 16 = 36$ gives $d = 5$ (1); dates 5, 6, 12, 13 (1); sum $= 4d + 16$ for any d (1).	1+1+1
	OR: Garden − pond $= 1200 - 80 = 1120$ m² (1); path $= (40+4)(30+4) - 1200 = 296$ m² (2).	1+2
25	$E = 3F/2 = 60$ (1); $V = 2 + E - F = 22$ (1); $F + V - E = 40 + 22 - 60 = 2$ (1).	1+1+1
	OR: New sum $= 1200 - 20 + 70 = 1250$ (2); mean $= 50$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $\frac{98}{7} = 14$ (1). (ii) Ordered: 9, 10, 12, 15, 15, 18, 19; median $= 15$ (1). (iii) Mode $= 15$ (1). (iv) Range $= 19 - 9 = 10$ (1); new mean $= \frac{98 + 20}{8} = 14.75$ (1).	1+1+1+2
27	$(100a + 10b + c) - (100c + 10b + a) = 99a - 99c$ (2); $= 99(a - c) = 99 \times 1$ (1); difference $= 99$ (2).	2+1+2

	OR: Trapezium $= \frac{1}{2}(14 + 26) \times 10 = 200$ cm² (2); parallelogram $= 140$ cm² (1); triangle $= \frac{1}{2} \times 12 \times 10 = 60$ cm² (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 4	$9(a-b)=9d$
2	**(b)** 405 cm²	$l+b=54$, $l=5b$: $b=9$, $l=45$
3	**(a)** 12	$V=2E/3=20$; $F=2+E-V$
4	**(d)** 26	Correct sum $=600-40+90$
5	**(c)** 14	$(x+1/x)^2=x^2+2+1/x^2$
6	**(c)** 204 cm²	Height $=\sqrt{13^2-5^2}=12$ cm; area $=\frac{1}{2}(12+22)\times 12$
7	**(d)** 14	6 original faces + 8 new triangular faces
8	**(a)** 37	$(24+1)(13)-24\times 12$
9	**(d)** 12	$4d+16$ = sum
10	**(c)** 2000	$2000\times 125/100\times 80/100$
11	**(c)** 5	$6(n-2)^2=54$ gives $n=5$
12	**(c)** 15	Sum of first n odd numbers is n^2, so mean $=n$
13	**(b)** Both A and R are true but R is not the correct explanation of A.	$3(4)+5=17$ so A is true; R is also true but it does not explain the value.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	A is true by $\frac{1}{2}\times 10\times 8=40$; R is also true but is a separate fact, so it does not explain A.
15	**(c)** A is true but R is false.	Edges $=3\times 6=18$ so A is true; a hexagonal prism has 12 vertices, so R is false.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Identity $(n+3)^2-(n-3)^2=12n$ (1); $=12\times 5=60$ (1).	1+1
17	$h=2A/b=15$ cm (1); new area $=270$ cm² (1).	1+1
18	$12(n-2)=84$ gives $n=9$ (1); total $=9^3=729$ (1).	1+1
19	Total $=6\times 35=210$ (1); rest $=5\times 32=160$, removed $=50$ (1).	1+1
20	$a+b=11$, $a-b=5$ (1); $a=8$, $b=3$, number 83 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$11(a+b)=132$ so $a+b=12$ (1); $a-b=2$ gives $a=7$, $b=5$ (1); number $=75$ (1).	1+1+1
22	$b=18$ m, $l=72$ m (1); area $=1296$ m² (1); fencing cost $=180\times 12=$ ₹ 2160 (1).	1+1+1
23	$12(n-2)=168$ gives $n=16$ (1); one face: $6(n-2)^2=6\times 196=1176$ (2).	1+2
24	Total age of players $=300$ (1); total with coach $=352$ (1); coach $=52$ years (1).	1+1+1
	OR: $x^2+1/x^2=4^2-2=14$ (1); $x^4+1/x^4=(14)^2-2=194$ (2).	1+2
25	Height $=\sqrt{20^2-12^2}=16$ cm (2); area $=\frac{1}{2}\times(20+44)\times 16=512$ cm² (1).	2+1
	OR: $3^6=729\le 1000$ but $3^7=2187>1000$ (2); so stage 7 with 2187 triangles (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $6n$ (1). (ii) $6n+12$ (1). (iii) $\dfrac{6n+12}{3}=2n+4$ (1). (iv) Subtract $2n$: $2n+4-2n=4$, independent of n (1); for $n=7$: $42, 54, 18, 18-14=4$ (1).	1+1+1+2
27	Using $(p+q)^2-(p-q)^2=4pq$ (2); $4\times 3n\times 2=24n$ (1); for $n=5$: 120 (2).	2+1+2
	OR: Other diagonal $=2\times 384/24=32$ cm (2); side $=\sqrt{12^2+16^2}=20$ cm (1); altitude $=384/20=19.2$ cm (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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